

Empowering Smart Glasses with Large Language Models: Towards Ubiquitous AGI

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Abstract

Smart glasses, augmented by advances in multimodal Large Language Models (LLMs), are at the forefront of creating ubiquitous Artificial General Intelligence (AGI). This short literature survey reviews the latest developments in integrating LLMs with smart glasses, emphasizing the technological enhancements, the opportunities, and the challenges faced. In addition to covering recent studies, it also proposes future research directions to harness the full potential of this emerging technology.

CCS Concepts

• **Human-centered computing** → Ubiquitous and mobile devices; Mixed / augmented reality; Natural language interfaces; • **Information systems** → Language models; Multimedia and multimodal retrieval; • **Computing methodologies** → Intelligent agents.

Keywords

Smart Glasses; Ubiquitous Computing; Augmented Reality; Large Language Models; Multimodal Interaction

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1 Introduction

The integration of Large Language Models (LLMs) [6, 10, 11] with smart glasses (such as Ray-Ban Meta¹ and Apple Vision Pro²) is revolutionizing the field of wearable technology. By combining the natural language processing capabilities of LLMs with the Augmented Reality (AR) features of smart glasses, these devices are moving towards providing more immersive, intelligent, and context-aware experiences. This short literature survey explores the latest advancements in this domain, highlighting important research contributions and identifying critical research challenges.

2 Technological Advancements

Recent research has focused on enhancing the capabilities of smart glasses through the incorporation of LLMs, enabling more natural and intuitive user interactions.

2.1 Multimodal Interaction

Multimodal interaction refers to the ability of devices to process and respond to multiple forms of input, such as voice, gesture, and visual data. The integration of newly emerged multimodal LLMs, like OpenAI GPT-4o³ and Google Gemini⁴, into smart glasses facilitates this by allowing sophisticated understanding and generation of language in conjunction with other input/output modalities.

For example, Yousri et al. [19] introduced IllusionX, a Mixed Reality (MR) system powered by LLMs to provide personalized educational companions. This system combines voice commands with AR displays to create an immersive learning environment. Similarly, Konrad et al. [8] developed GazeGPT, which utilizes gaze direction as an input for smart glasses to augment human capabilities, demonstrating the potential of multimodal interaction in enhancing user experience. Systems like PANDALens [2] leverage multimodal contextual information to transform in-context writing experiences, making them more intuitive and less effortful.

¹<https://www.meta.com/smart-glasses/>

²<https://www.apple.com/apple-vision-pro/>

³<https://openai.com/index/hello-gpt-4o/>

⁴<https://deepmind.google/technologies/gemini/>

2.2 Context-Aware Systems

Context-awareness is crucial for the effectiveness of smart glasses, enabling them to adapt to the user's environment and provide relevant information and assistance.

Iqbal and Campbell discussed the ethical and privacy implications of context-aware smart glasses, particularly focusing on devices like Ray-Ban Stories by Meta [7]. Their work highlights the balance between technological advancement and privacy concerns, which is essential for the widespread adoption of such technologies. Furthermore, VisionARY [9] integrates contextual language learning into AR glasses, enhancing the user's learning environment by providing real-time, relevant information.

2.3 Extended Reality

The use of AR or MR in smart glasses has been a significant focus, with efforts to improve the realism and interactivity of virtual elements.

Constantinides et al. [4] explored the integration of LLMs within MR environments for virtual art exhibitions, enhancing user interaction through voice-based inquiries about artworks. This integration not only makes the interaction more engaging but also leverages the deep learning capabilities of LLMs to provide detailed and accurate responses. RealitySummary [5] enhances document reading by using LLMs for on-demand text extraction, summarization, and augmentation, improving user engagement with physical and digital texts.

3 Applications and Opportunities

The applications of smart glasses equipped with LLMs span various domains, including education, healthcare, and daily personal assistance to name a few.

3.1 Educational Tools

Educational applications are among the most promising uses of this technology. IllusionX [19], for instance, provides a comprehensive system for personal learning and teaching assistance, leveraging LLMs to generate detailed educational content and assist with interactive lessons. Similarly, VisionARY [9] integrates ChatGPT into AR glasses to improve language learning experiences by providing contextual language learning support. CASES [15] combines semantic context and visual attention to better understand cognitive reading states, demonstrating potential applications in cognitive and educational fields.

3.2 Healthcare Assistance

In healthcare, smart glasses can provide real-time assistance to medical professionals and patients. For instance, Plizzari et al. [14] reviewed the future of egocentric vision in healthcare, highlighting the potential of smart glasses to aid in surgical procedures and patient monitoring. Additionally, EMOShip enhances emotion recognition by analyzing associations with visual perceptions, providing valuable insights for healthcare applications [20]. TEXT2TASTE [13] has been developed to help individuals with (partial) vision impairments by leveraging LLMs to read and summarize text, enhancing

independence and daily life activities. Waisberg et al. [18] also discussed how LLMs could enhance Ray-Ban Meta smart glasses for assisting individuals with vision impairments.

3.3 Daily Life Enhancement

For everyday use, systems like GazeNoter [17] demonstrate how smart glasses can assist with tasks such as note-taking through gaze selection, significantly improving productivity and convenience. Furthermore, GlassMail [21] provides a personalized wearable assistant for on-the-go email creation, showcasing the potential for smart glasses in everyday scenarios.

4 Challenges and Directions

Despite the advancements, several critical challenges remain in the widespread use of smart glasses powered by LLMs. These include privacy/ethical concerns and the need for high-performance hardware that ensures user comfort over extended periods.

4.1 Privacy and Ethics

Privacy has always been a major concern, especially with the potential for constant recording and data collection by smart glasses. Researchers like Iqbal and Campbell [7] emphasize the importance of addressing these issues to gain public trust and acceptance. Bozkir et al. [1] also discuss the potential privacy invasions and ethical considerations of LLM-powered AR/MR environments.

4.2 Hardware Limitations

The hardware of smart glasses must evolve to support the computational demands of LLMs without compromising on battery life or user comfort. Lightweight, durable designs that can house powerful processors and batteries are essential for the practical deployment of these devices. Advances like MemX [3], which enhances energy efficiency and attention tracking accuracy, are steps towards addressing these limitations.

4.3 Future Research

Future research need to focus on improving the egocentric [16] multimodal understanding and generation capabilities of edge LLMs in the context of smart glasses, developing more efficient hardware, and addressing privacy as well as ethical issues comprehensively. Additionally, exploring new applications in diverse fields can further demonstrate the versatility and utility of this technology. The survey paper from Ma et al. [12] and Plizzari et al. [14] provide comprehensive overviews of current methodologies and can guide future research in developing more advanced vision-language-action models for smart glasses.

5 Conclusion

The integration of (multimodal) LLMs with smart glasses marks a significant step towards ubiquitous AGI. This short literature survey has highlighted the recent advancements and applications of this technology, as well as the challenges that need to be addressed. Continued research and development in this area hold the promise of transforming how we interact with both the physical world and the digital world through smart glasses.

References

- [1] Efe Bozkir, Süleyman Özdel, Ka Hei Carrie Lau, Mengdi Wang, Hong Gao, and Enkelejda Kasneci. 2024. Embedding Large Language Models into Extended Reality: Opportunities and Challenges for Inclusion, Engagement, and Privacy. <https://doi.org/10.1145/3640794.3665563> arXiv:2402.03907 [cs]
- [2] Runze Cai, Nuwan Janaka, Yang Chen, Lucia Wang, Shengdong Zhao, and Can Liu. 2024. PANDALens: Towards AI-Assisted In-Context Writing on OHMD During Travels. In *Proceedings of the CHI Conference on Human Factors in Computing Systems (CHI '24)*. Association for Computing Machinery, New York, NY, USA, 1–24. <https://doi.org/10.1145/3613904.3642320>
- [3] Yuhu Chang, Yingying Zhao, Mingzhi Dong, Yujiang Wang, Yutian Lu, Qin Lv, Robert P. Dick, Tun Lu, Ning Gu, and Li Shang. 2021. MemX: An Attention-Aware Smart Eyewear System for Personalized Moment Auto-capture. *Proc. ACM Interact. Mob. Wearable Ubiquitous Technol.* 5, 2 (June 2021), 56:1–56:23. <https://doi.org/10.1145/3463509>
- [4] Nicolas Constantinides, Argyris Constantinides, Dimitrios Koukopoulos, Christos Fidas, and Marios Belk. 2024. CulturAI: Exploring Mixed Reality Art Exhibitions with Large Language Models for Personalized Immersive Experiences. In *Adjunct Proceedings of the 32nd ACM Conference on User Modeling, Adaptation and Personalization*. ACM, Cagliari Italy, 102–105. <https://doi.org/10.1145/3631700.3664874>
- [5] Aditya Gunturu, Shivesh Jadon, Nandi Zhang, Jarin Thundathil, Wesley Willett, and Ryo Suzuki. 2024. RealitySummary: On-Demand Mixed Reality Document Enhancement Using Large Language Models. <https://doi.org/10.48550/arXiv.2405.18620> arXiv:2405.18620 [cs]
- [6] Zhongjiang He, Zihan Wang, Xinzhang Liu, Shixuan Liu, Yitong Yao, Yuyao Huang, Xuelong Li, Yongxiang Li, Zhonghao Che, Zhaoxi Zhang, Yan Wang, Xin Wang, Luwen Pu, Huinan Xu, Ruiyu Fang, Yu Zhao, Jie Zhang, Xiaomeng Huang, Zhilong Lu, Jiaxin Peng, Wenjun Zheng, Shiquan Wang, Bingkai Yang, Xuewei he, Zhuoru Jiang, Qiyi Xie, Yanhan Zhang, Zhongqiu Li, Lingling Shi, Weiwei Fu, Yin Zhang, Zilu Huang, Sishi Xiong, Yuxiang Zhang, Chao Wang, and Shuangyong Song. 2024. TeleChat Technical Report. <https://doi.org/10.48550/arXiv.2401.03804> arXiv:2401.03804 [cs]
- [7] Muhammad Zahid Iqbal and Abraham G. Campbell. 2023. Adopting Smart Glasses Responsibly: Potential Benefits, Ethical, and Privacy Concerns with Ray-Ban Stories. *AI and Ethics* 3, 1 (Feb. 2023), 325–327. <https://doi.org/10.1007/s43681-022-00155-7>
- [8] Robert Konrad, Nitish Padmanaban, J. Gabriel Buckmaster, Kevin C. Boyle, and Gordon Wetzstein. 2024. GazeGPT: Augmenting Human Capabilities Using Gaze-contingent Contextual AI for Smart Eyewear. <https://doi.org/10.48550/arXiv.2401.17217> arXiv:2401.17217 [cs]
- [9] Hyungmin Lee, Chen-Chun Hsia, Aleksandr Tsoy, Sungmin Choi, Hanchao Hou, and Shiguang Ni. 2023. VisionARY: Exploratory Research on Contextual Language Learning Using AR Glasses with ChatGPT. In *Proceedings of the 15th Biannual Conference of the Italian SIGCHI Chapter (CHIItaly '23)*. Association for Computing Machinery, New York, NY, USA, 1–6. <https://doi.org/10.1145/3605390.3605400>
- [10] Xiang Li, Yiqun Yao, Xin Jiang, Xuezhong Fang, Chao Wang, Xinzhang Liu, Zihan Wang, Yu Zhao, Xin Wang, Yuyao Huang, Shuangyong Song, Yongxiang Li, Zheng Zhang, Bo Zhao, Aixin Sun, Yequan Wang, Zhongjiang He, Zhongyuan Wang, Xuelong Li, and Tiejun Huang. 2024. 52B to 1T: Lessons Learned via Tele-FLM Series. <https://doi.org/10.48550/arXiv.2407.02783> arXiv:2407.02783 [cs]
- [11] Xiang Li, Yiqun Yao, Xin Jiang, Xuezhong Fang, Chao Wang, Xinzhang Liu, Zihan Wang, Yu Zhao, Xin Wang, Yuyao Huang, Shuangyong Song, Yongxiang Li, Zheng Zhang, Bo Zhao, Aixin Sun, Yequan Wang, Zhongjiang He, Zhongyuan Wang, Xuelong Li, and Tiejun Huang. 2024. Tele-FLM Technical Report. <https://doi.org/10.48550/arXiv.2404.16645> arXiv:2404.16645 [cs]
- [12] Yuen Ma, Zixing Song, Yuzheng Zhuang, Jianye Hao, and Irwin King. 2024. A Survey on Vision-Language-Action Models for Embodied AI. <https://doi.org/10.48550/arXiv.2405.14093> arXiv:2405.14093 [cs]
- [13] Wiktor Mucha, Florin Cuconasu, Naome A. Etori, Valia Kalokyri, and Giovanni Trappolini. 2024. TEXT2TASTE: A Versatile Egocentric Vision System for Intelligent Reading Assistance Using Large Language Model. <https://doi.org/10.48550/arXiv.2404.09254> arXiv:2404.09254 [cs]
- [14] Chiara Plizzari, Gabriele Goletto, Antonino Furnari, Siddhant Bansal, Francesco Ragusa, Giovanni Maria Farinella, Dima Damen, and Tatiana Tommasi. 2024. An Outlook into the Future of Egocentric Vision. <https://doi.org/10.48550/arXiv.2308.07123> arXiv:2308.07123 [cs]
- [15] Xiangyao Qi, Qi Lu, Wentao Pan, Yingying Zhao, Rui Zhu, Mingzhi Dong, Yuhu Chang, Qin Lv, Robert P. Dick, Fan Yang, Tun Lu, Ning Gu, and Li Shang. 2023. CASES: A Cognition-Aware Smart Eyewear System for Understanding How People Read. *Proc. ACM Interact. Mob. Wearable Ubiquitous Technol.* 7, 3 (Sept. 2023), 115:1–115:31. <https://doi.org/10.1145/3610910>
- [16] Alessandro Suglia, Claudio Greco, Katie Baker, Jose L. Part, Ioannis Papaioannou, Arash Eshghi, Ioannis Konstantas, and Oliver Lemon. 2024. AlanaVLM: A Multimodal Embodied AI Foundation Model for Egocentric Video Understanding. <https://doi.org/10.48550/arXiv.2406.13807> arXiv:2406.13807 [cs]
- [17] Hsin-Ruey Tsai, Shih-Kang Chiu, and Bryan Wang. 2024. GazeNoter: Co-Piloted AR Note-Taking via Gaze Selection of LLM Suggestions to Match Users' Intentions. <https://doi.org/10.48550/arXiv.2407.01161> arXiv:2407.01161 [cs]
- [18] Ethan Waisberg, Joshua Ong, Mouayad Masalkhi, Nasif Zaman, Prithul Sarker, Andrew G. Lee, and Alireza Tavakkoli. 2024. Meta Smart Glasses — Large Language Models and the Future for Assistive Glasses for Individuals with Vision Impairments. *Eye* 38, 6 (April 2024), 1036–1038. <https://doi.org/10.1038/s41433-023-02842-z>
- [19] Ramez Yousri, Zeyad Essam, Yehia Kareem, Youstina Sherief, Sherry Gamil, and Soha Safwat. 2024. IllusionX: An LLM-Powered Mixed Reality Personal Companion. <https://doi.org/10.48550/arXiv.2402.07924> arXiv:2402.07924 [cs]
- [20] Yingying Zhao, Yuhu Chang, Yutian Lu, Yujiang Wang, Mingzhi Dong, Qin Lv, Robert P. Dick, Fan Yang, Tun Lu, Ning Gu, and Li Shang. 2022. Do Smart Glasses Dream of Sentimental Visions? Deep Emotionship Analysis for Eyewear Devices. *Proc. ACM Interact. Mob. Wearable Ubiquitous Technol.* 6, 1 (March 2022), 38:1–38:29. <https://doi.org/10.1145/3517250>
- [21] Chen Zhou, Zihan Yan, Ashwin Ram, Yue Gu, Yan Xiang, Can Liu, Yun Huang, Wei Tsang Ooi, and Shengdong Zhao. 2024. GlassMail: Towards Personalised Wearable Assistant for On-the-Go Email Creation on Smart Glasses. In *Designing Interactive Systems Conference*. ACM, IT University of Copenhagen Denmark, 372–390. <https://doi.org/10.1145/3643834.3660683>